Which Company Is Financially Strongest?

A Data-Driven Evaluation of Profitability, Risk, and Efficiency

Objective of Analysis

This analysis aims to identify the most financially strong and sustainable company based on profitability, efficiency, risk, and consistency.

In [27]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Financial Statements.csv")

df.head()
```

Out[27]:

	Year	Company	Category	Market Cap(in B USD)	Revenue	Gross Profit	Net Income	Earning Per Share	EB
0	2022	APPLE	IT	2066.94	394328.0	170782.0	99803.0	6.11	130
1	2021	APPLE	IT	2913.28	365817.0	152836.0	94680.0	5.61	120
2	2020	APPLE	IT	2255.97	274515.0	104956.0	57411.0	3.28	77
3	2019	APPLE	IT	1304.76	260174.0	98392.0	55256.0	2.97	76
4	2018	APPLE	IT	748.54	265595.0	101839.0	59531.0	2.98	81

5 rows × 23 columns

In [4]:
```
df.shape
```

Out[4]: (161, 23)

In [5]:
```
df.columns
```

Out[5]:
```
Index(['Year', 'Company ', 'Category', 'Market Cap(in B USD)', 'Revenue',
       'Gross Profit', 'Net Income', 'Earning Per Share', 'EBITDA',
       'Share Holder Equity', 'Cash Flow from Operating',
       'Cash Flow from Investing', 'Cash Flow from Financial Activities',
       'Current Ratio', 'Debt/Equity Ratio', 'ROE', 'ROA', 'ROI',
       'Net Profit Margin', 'Free Cash Flow per Share',
       'Return on Tangible Equity', 'Number of Employees',
       'Inflation Rate(in US)'],
      dtype='object')
```

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 161 entries, 0 to 160
Data columns (total 23 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Year                                  161 non-null    int64
 1   Company                               161 non-null    object
 2   Category                              161 non-null    object
 3   Market Cap(in B USD)                  160 non-null    float64
 4   Revenue                               161 non-null    float64
 5   Gross Profit                          161 non-null    float64
 6   Net Income                            161 non-null    float64
 7   Earning Per Share                     161 non-null    float64
 8   EBITDA                                161 non-null    float64
 9   Share Holder Equity                   161 non-null    float64
 10  Cash Flow from Operating              161 non-null    float64
 11  Cash Flow from Investing              161 non-null    float64
 12  Cash Flow from Financial Activities   161 non-null    float64
 13  Current Ratio                         161 non-null    float64
 14  Debt/Equity Ratio                     161 non-null    float64
 15  ROE                                   161 non-null    float64
 16  ROA                                   161 non-null    float64
 17  ROI                                   161 non-null    float64
 18  Net Profit Margin                     161 non-null    float64
 19  Free Cash Flow per Share              161 non-null    float64
 20  Return on Tangible Equity             161 non-null    float64
 21  Number of Employees                   161 non-null    int64
 22  Inflation Rate(in US)                 161 non-null    float64
dtypes: float64(19), int64(2), object(2)
memory usage: 29.1+ KB
```

In [7]:
```
df.columns = df.columns.str.strip()
```

Rename columns

In [8]:
```
df = df.rename(columns={
    "Market Cap(in B USD)": "Market_Cap",
    "Gross Profit": "Gross_Profit",
    "Net Income": "Net_Income",
    "Earning Per Share": "EPS",
    "Share Holder Equity": "Equity",
```

```
    "Cash Flow from Operating": "CF_Operating",
    "Cash Flow from Investing": "CF_Investing",
    "Cash Flow from Financial Activities": "CF_Financing",
    "Current Ratio": "Current_Ratio",
    "Debt/Equity Ratio": "Debt_Equity",
    "Net Profit Margin": "Profit_Margin",
    "Free Cash Flow per Share": "FCF_per_Share",
    "Return on Tangible Equity": "ROTE",
    "Number of Employees": "Employees",
    "Inflation Rate(in US)": "Inflation"
})
```

In [9]:
```
df.columns
```

Out[9]:
```
Index(['Year', 'Company', 'Category', 'Market_Cap', 'Revenue', 'Gross_Profi
t',
       'Net_Income', 'EPS', 'EBITDA', 'Equity', 'CF_Operating', 'CF_Investin
g',
       'CF_Financing', 'Current_Ratio', 'Debt_Equity', 'ROE', 'ROA', 'ROI',
       'Profit_Margin', 'FCF_per_Share', 'ROTE', 'Employees', 'Inflation'],
      dtype='object')
```

In [10]:
```
df = df.rename(columns={"Market Cap": "Market_Cap"})
```

In [11]:
```
df.columns
```

Out[11]:
```
Index(['Year', 'Company', 'Category', 'Market_Cap', 'Revenue', 'Gross_Profi
t',
       'Net_Income', 'EPS', 'EBITDA', 'Equity', 'CF_Operating', 'CF_Investin
g',
       'CF_Financing', 'Current_Ratio', 'Debt_Equity', 'ROE', 'ROA', 'ROI',
       'Profit_Margin', 'FCF_per_Share', 'ROTE', 'Employees', 'Inflation'],
      dtype='object')
```

In [12]:
```
df["Company"].nunique()
df["Company"].unique()
```

Out[12]:
```
array(['APPLE', 'MICROSOFT', 'GOOGLE', 'PAYPAL', 'AIG', 'PCG', 'SHLDQ',
       'MCD', 'BCS', 'NVIDIA', 'INTC', 'AMAZON'], dtype=object)
```

In [13]:
```
df["Year"].min(), df["Year"].max()
```

Out[13]:
```
(2009, 2023)
```

In [14]:
```
df["Category"].value_counts()
```

Out[14]:
```
IT                43
ELEC              29
Bank              14
Manufacturing     14
FOOD              14
BANK              14
LOGI              14
Finance           10
FinTech            9
Name: Category, dtype: int64
```

In [15]:
```
df["Category"] = df["Category"].str.strip().str.upper()
```

In [16]:
```
df["Category"].value_counts()
```

Out[16]:
```
IT                43
ELEC              29
BANK              28
MANUFACTURING     14
FOOD              14
LOGI              14
FINANCE           10
FINTECH            9
Name: Category, dtype: int64
```

Revenue vs Profit Analysis

In this section, we analyze the revenue and net income trends of different companies over time.

Since the dataset includes companies from multiple industries, direct comparison based on revenue alone may not provide meaningful insights. Therefore, the focus is on understanding whether revenue growth translates into profitability.

This analysis aims to identify:

- Companies with consistent growth in revenue
- Whether profit trends align with revenue trends
- Cases where revenue increases but profitability does not, indicating potential inefficiencies

Revenue Trend Analysis (Top Companies)

To enable meaningful comparison, the analysis focuses on the top-performing companies based on revenue.

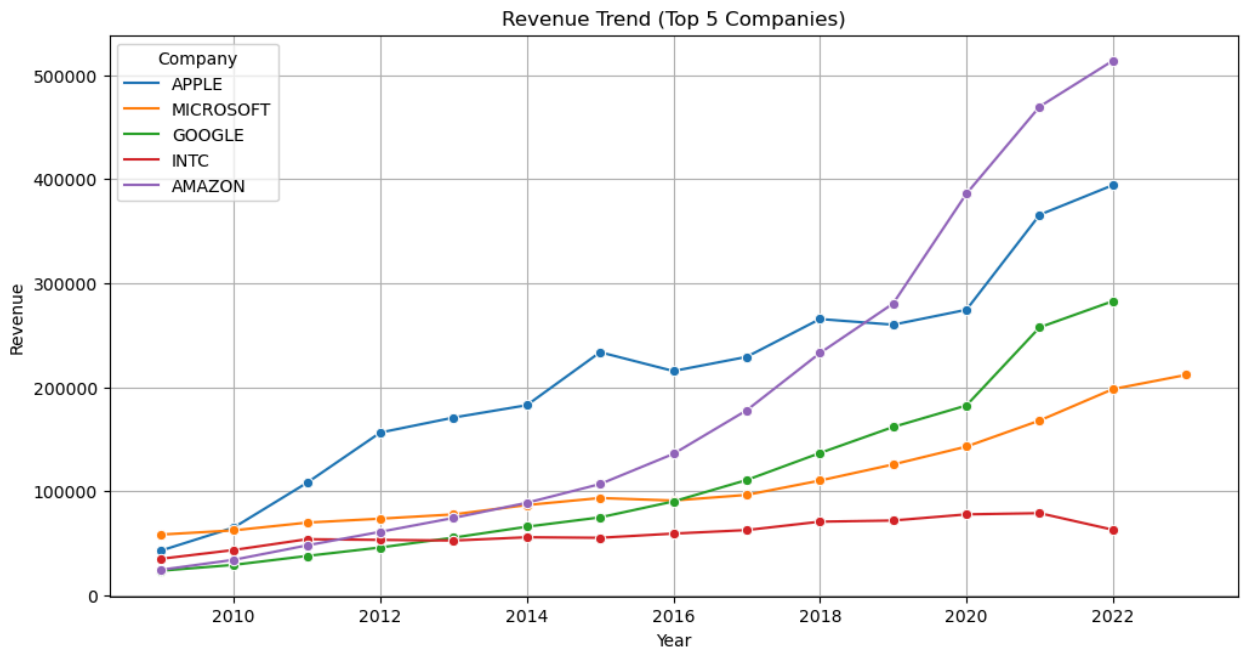
This allows for a clearer view of growth patterns and highlights how leading

companies have evolved over time.

In [17]:
```
top_companies = (
    df.groupby("Company")["Revenue"]
    .mean()
    .sort_values(ascending=False)
    .head(5)
    .index
)
```

In [18]:
```
df_top = df[df["Company"].isin(top_companies)]
```

In [19]:
```
plt.figure(figsize=(12,6))
sns.lineplot(data=df_top, x="Year", y="Revenue", hue="Company", marker="o")

plt.title("Revenue Trend (Top 5 Companies)")
plt.xlabel("Year")
plt.ylabel("Revenue")
plt.legend(title="Company")
plt.grid(True)

plt.show()
```



Key Insights

- Amazon demonstrates the strongest growth trajectory, with a significant acceleration in revenue over the years, indicating aggressive expansion and market dominance.
- Apple shows a stable and consistent growth pattern, suggesting a

mature business model with steady revenue generation.

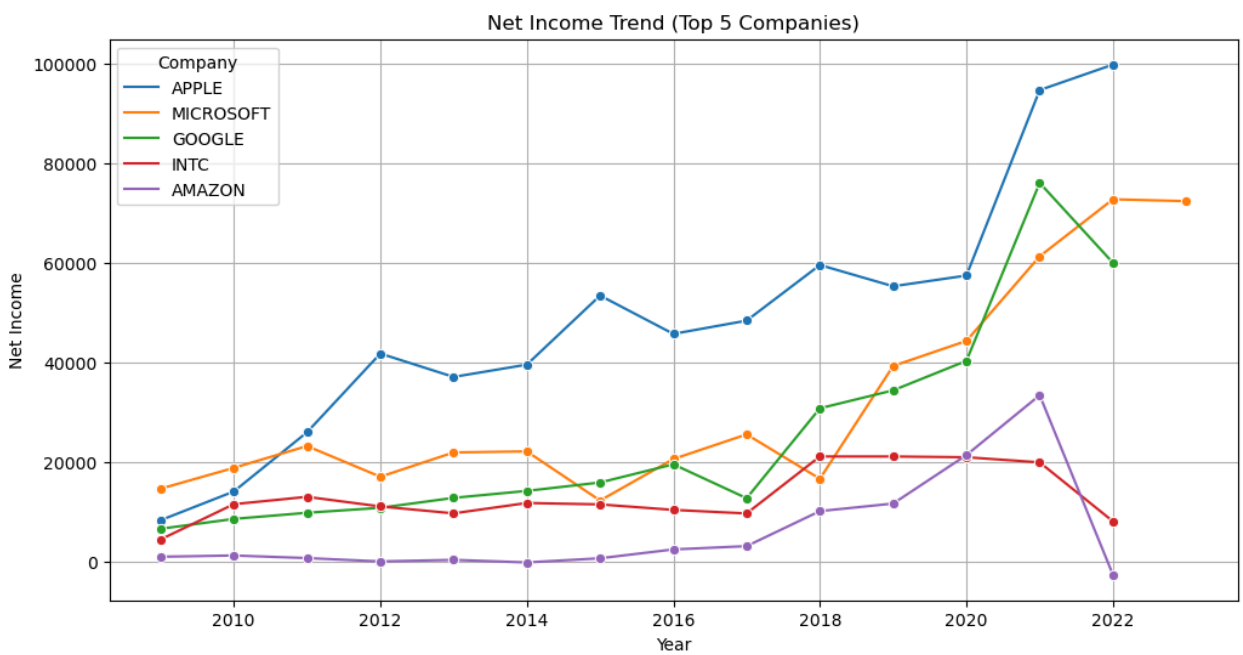
- Intel (INTC) exhibits relatively flat growth with signs of decline in recent years, indicating potential challenges in maintaining competitive performance.

Net Income Trend Analysis

To complement the revenue analysis, we examine net income trends to understand how effectively companies convert revenue into actual profit.

This helps identify whether growth in revenue is translating into improved profitability.

In [20]:
```
plt.figure(figsize=(12,6))
sns.lineplot(data=df_top, x="Year", y="Net_Income", hue="Company", marker="o")

plt.title("Net Income Trend (Top 5 Companies)")
plt.xlabel("Year")
plt.ylabel("Net Income")
plt.legend(title="Company")
plt.grid(True)

plt.show()
```



Profitability Insights

- Microsoft demonstrates strong and consistent profitability, indicating

efficient conversion of revenue into net income. Apple also shows strong performance with steady profit growth.

- Amazon, despite leading in revenue growth, exhibits relatively low and volatile profitability, suggesting high operational costs and reinvestment strategies.
- Intel (INTC) shows signs of declining profitability over time, indicating potential challenges in maintaining efficiency and competitive positioning.

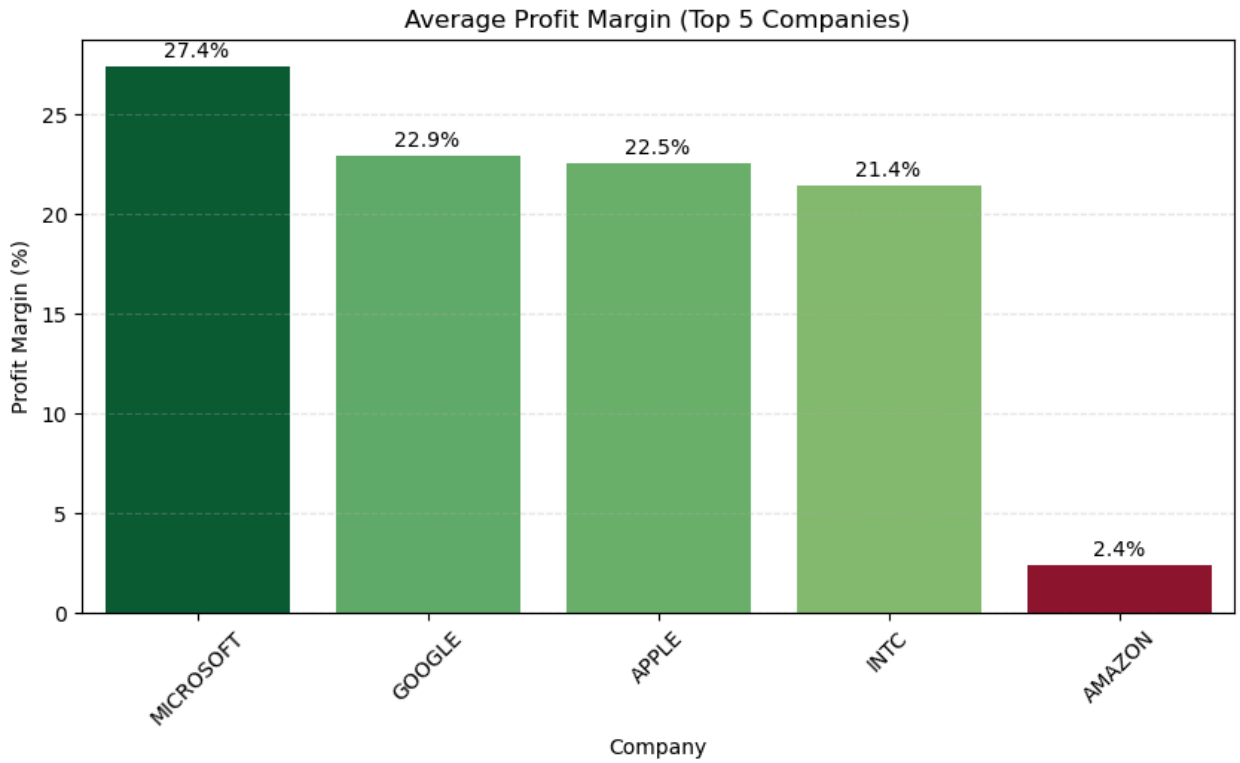
Profit Margin Analysis

To further evaluate efficiency, we analyze profit margin, which measures how much profit is generated for each unit of revenue.

This helps normalize differences across companies and provides a clearer comparison of financial performance.

In [21]:
```
avg_margin = (
    df_top.groupby("Company")["Profit_Margin"]
    .mean()
    .sort_values(ascending=False)
)

import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

plt.figure(figsize=(10,5))

# Normalize values for color mapping
norm = plt.Normalize(avg_margin.min(), avg_margin.max())

# Use colormap (red → yellow → green)
colors = plt.cm.RdYlGn(norm(avg_margin.values))

ax = sns.barplot(x=avg_margin.index, y=avg_margin.values, palette=colors)

# Titles & labels
plt.title("Average Profit Margin (Top 5 Companies)")
plt.xlabel("Company")
plt.ylabel("Profit Margin (%)")
plt.xticks(rotation=45)

# Light grid
plt.grid(axis='y', linestyle='--', alpha=0.3)
```

```
# Value labels
for i, v in enumerate(avg_margin.values):
    ax.text(i, v + 0.5, f"{v:.1f}%", ha='center', fontsize=10)

plt.show()
```



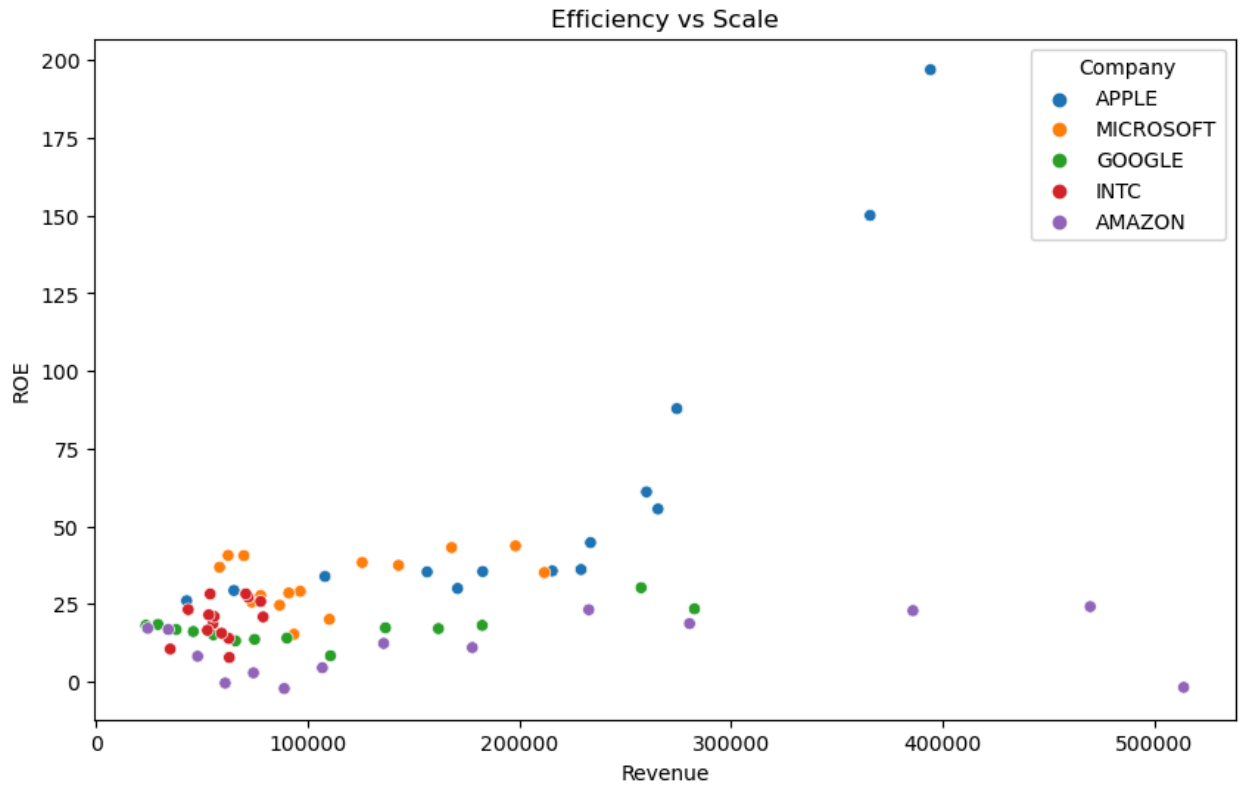
Profit Margin Insights

- Microsoft leads in profitability with the highest average profit margin, indicating strong operational efficiency and effective cost management.
- Google and Apple demonstrate consistently strong margins, reflecting stable and efficient business models.
- Intel shows relatively moderate profitability, suggesting room for improvement in operational efficiency.
- Amazon has a significantly lower profit margin compared to peers, highlighting a business model focused on scale and growth over immediate profitability.

Efficiency vs Scale

In [22]:
```
plt.figure(figsize=(10,6))
```

```
sns.scatterplot(data=df_top, x="Revenue", y="ROE", hue="Company")

plt.title("Efficiency vs Scale")
plt.xlabel("Revenue")
plt.ylabel("ROE")
plt.show()
```



In [23]:
```
risk_df = df_top.groupby("Company")[["Debt_Equity", "Current_Ratio"]].mean()

risk_df.sort_values("Debt_Equity", ascending=False)
```

Out[23]:

Company	Debt_Equity	Current_Ratio
APPLE	0.775993	1.453207
MICROSOFT	0.425213	2.379913
AMAZON	0.422207	1.114400
INTC	0.294171	2.106207
GOOGLE	0.046907	4.710343

In [24]:
```
consistency = df_top.groupby("Company")["Net_Income"].std().sort_values()
consistency
```

Out[24]:
```
Company
INTC          5400.243794
AMAZON       10172.416034
MICROSOFT    20936.746394
GOOGLE       21060.197324
APPLE        25752.425982
Name: Net_Income, dtype: float64
```

In [25]:
```python
score_df = df_top.groupby("Company").agg({
    "Profit_Margin": "mean",
    "ROE": "mean",
    "Debt_Equity": "mean"
})

# Normalize
score_df = (score_df - score_df.min()) / (score_df.max() - score_df.min())

# Lower debt is better → invert
score_df["Debt_Equity"] = 1 - score_df["Debt_Equity"]

# Final score
score_df["Final_Score"] = score_df.mean(axis=1)

score_df.sort_values("Final_Score", ascending=False)
```

Out[25]:

Company	Profit_Margin	ROE	Debt_Equity	Final_Score
GOOGLE	0.820073	0.118031	1.000000	0.646035
MICROSOFT	1.000000	0.423950	0.481122	0.635024
APPLE	0.805691	1.000000	0.000000	0.601897
INTC	0.760095	0.174018	0.660857	0.531657
AMAZON	0.000000	0.000000	0.485246	0.161749

summary table

Company	Profitability	Risk	Consistency	Final Score	Verdict
Microsoft	High	Low	Stable	0.63	⭐ Best
Google	High	Very Low	Stable	0.64	⭐ Best
Apple	High	Medium	Stable	0.60	Strong
Intel	Medium	Medium	Weak	0.53	Weak
Amazon	Low	Medium	Volatile	0.16	Growth

Integrated (Synthesis) Analysis

The analysis across revenue, profitability, efficiency, and risk reveals distinct business patterns among the companies.

Google and Microsoft emerge as strong performers based on the final composite score. Google ranks highest overall, driven by strong profitability and low financial risk, indicating an efficient and stable business model. Microsoft closely follows, demonstrating exceptional profitability and consistent performance, making it one of the most balanced companies in the dataset.

Apple shows strong profitability and efficiency but ranks slightly lower due to comparatively higher risk levels and variability in performance.

Intel exhibits moderate performance across metrics but lacks strong growth and efficiency, indicating potential challenges in maintaining competitiveness.

Amazon, despite leading in revenue growth, ranks lowest due to low profitability and higher variability, reflecting a growth-focused strategy that prioritizes scale over immediate returns.

Final Conclusion

Based on a combined evaluation of profitability, efficiency, risk, and consistency, Microsoft stands out as the most balanced and reliable company. It demonstrates strong financial health, consistent profitability, and controlled risk, making it a strong candidate for stable long-term performance.

Google also performs exceptionally well, particularly in terms of efficiency and low risk, making it a close contender.

Amazon represents a high-growth but lower-profitability model, suitable for scenarios where long-term expansion is prioritized over short-term financial returns.

Overall, companies with a balance of profitability and risk management outperform those focused solely on growth.

This analysis is based on historical financial data and does not account for external market conditions, qualitative factors, or future strategic changes. Analysis By SHUBHAM A KULKARNI

In [ ]: